

Course Syllabus

Course name: Human Microscopic and Developmental Anatomy (Lectures)

Course Numbers: DENT 601A

Credit Hours: 3

Contact Hours: 42

Course Director: Dongmei Cui, MD (hon.), Ph.D. Professor, Department of Advanced Biomedical Education, University of Mississippi Medical Center. E-mail: dcui@umc.edu; Office: N-710A

Course Faculty:

Dongmei Cui, MD (hon), Ph.D. Email: dcui@umc.edu

Edgar R. Meyer, Ph.D. Email: emeyer@umc.edu

Teaching Assistant:

Jacob A Roberts, MS (Clinical Anatomy Ph.D. Student)

Email: jroberts11@umc.edu

Clinical Faculty:

William P. Daley, M.D.

Jamie A. White, D.M.D.

Jennifer L. Bain, D.M.D.

Joseph Bryant Holsch, D.M.D.

Course Description and Goals:

This is a 15 weeks course in study human microscopic (histology) and developmental anatomy, where first year dental students will gain knowledge and experiences in microscopic exam of histological slides, and study of the structure and function of cells, tissues and organs; especially structures inside of the oral cavity. This course will integrate some clinical correlations associated with basic tissues, organ system and oral structures.

This course consists of:

- 1) **Cell Biology:** It includes Cell membrane, Cell membrane transport role, Cell nucleus, Cytoplasm and Organelles (Mitochondria, Ribosome, Rough endoplasmic reticulum, Smooth endoplasmic reticulum, Golgi complex/ Golgi apparatus, Lysosome and Vesicles.

- 2) **Basic Tissues:** Epithelium and exocrine glands, Connective Tissue (CT), Cartilage and bone, Muscle Tissue (MT), Nervous Tissue (PNS, ANS and CNS).
- 3) **Organ system:** Circulatory System and Peripheral Blood, Immune system and Lymphoid Organs, Integument-Skin, Respiratory System, Urinary system, Endocrine System, Digestive Tract, Digestive Glands and Salivary glands.
- 4) **Oral Embryology:** Development of the Oral Facial Region, Development of the Face and Palate
- 5) **Oral Cavity:** Oral mucosal and Tongue, Odontogenesis, Odontogenesis, enamel, Dentin, Pulp, Periodontium (Cementum, Periodontal ligaments, alveolar bone).
- 6) **Clinical Correlations:** Organ system clinical correlations, Oral pathology, Osseo-integration and TMJ and clinical correlation.

Our goals are helping students to learn the fundamental of histology, cell biology, oral embryology and oral histology; hope students will strengthen their knowledge of basic biology of cells, tissues and organ system and their knowledge of oral cavity structures, cells; and tooth and facial development. Hopefully, students will be able to facilitate their ability to integrate microscopic anatomy in their dental disciplines.

Course Schedule: Details see DENT 601A&B Schedule

Monday 8am to 12pm

Wednesday 8am to 12 pm

Course Objectives:

- Be able to utilize the vocabulary of microscopic anatomy /histological sciences associated with cells, tissues, organs and structures of the oral cavity
- Ability to describe the structure and function of the cell membrane, nucleus, cytoplasm and organelles.
- Ability to describe histological features of each type of tissue (epithelium, connective tissue, cartilage and bone, muscle and nerve tissue-NPS, ANS and CNS)
- Be capable of describing the main function of each type of tissue.
- Be able to classify each basic tissue and its sub-type (if applicable)
- Ability to describe the histological features and main function for each organ and its associated structures and cells.
- Be capable of identifying the organs in each organ system and their associated structures and cells.
- Able to understand developments in the oral and facial region and tooth developments.
- To be able to describe the microscopic features and functions of oral structures and cells.
- Understanding of some clinical correlations associated with histological cells, structures, organs and oral cavity.

Assessment Method:

The student's grade will be based on four Exams and Bonus Quizzes.

Grading Scale

A=90 to 100

B= 80 to less than 90

C= 70 to less than 80

F= below 70

Exam 1: Cell Biology and Basic Tissue (25%), topics include Cell membrane, Nucleus, Cytoplasm, Organelles; Epithelium and Exocrine glands; Connective Tissue; Cartilage and Bone; Muscle Tissue; Nervous Tissue.

Exam 2: Peripheral Blood and Organ System one (25%), topics include Peripheral blood; Circulatory System; Immune System; Lymphoid Organs; Integument-Skin; Respiratory System.

Exam 3: Organ System two (25%), topics include Urinary System; Endocrine System; Digestive Tract; Digestive Glands and associated Organs, Salivary Glands and Embryology

Exam 4: Oral Cavity (25%), topics include Development of the Oral Facial Region; Development of the Face and Palate; Odontogenesis; Oral mucosa and Tongue; Enamel, Dentin and Dental Pulp; Periodontium (periodontal ligament, Cementum and Alveolar bone).

Bonus Quizzes (Optional): Students can earn bonus points via completion of bonus questions with correct answers. In general, the Quizzes assessment usually contains 10-15 questions of varied formats: matching, True/ False, multiple choice. Accumulated bonus points for each block will be added to the written score achieved on that block written exam.

Weighted Written grades calculation formula:

Block 1 Grade = [Written exam grade (Block 1) + Bonus Points] * 25%

Block 2 Grade = [Written exam grade (Block 2) + Bonus Points] * 25%

Block 3 Grade = [Written exam grade (Block 3) + Bonus Points] * 25%

Block 4 Grade = [Written exam grade (Block 4) + Bonus Points] * 25%

Final Grade = Block 1 grade (points) + Block 2 grade (points) + Block 3 grade (points) + Block 4 grade (points)

Each exam grade is calculated to 2 decimal points. The final course grade is entered into Workday. The max points are 100 for the final grade to be entered into Workday, the students who have over 100 with bonus points will be acknowledged to your Assistant Dean for Education, Dr. Lott. The successful student must achieve a final weighted grade of ≥ 70.00 to pass DENT 601A.

Educational Resources:

1. Textbooks: (Suggested book based on previous year Required texts)
Mescher, A. Junqueira' Basic Histology: Text and Atlas. 17th edition. McGraw-Hill Education. ISBN-13:987-0071780339

Cui et al. Histology from a Clinical Perspective. 2nd edition. Wolters Kluwer. ISBN: 978-1-9751-5244-4

Berkovitz BKB, Holland GR, Moxham BJ. Oral Anatomy, Histology and Embryology. 6th edition, Mosby. ISBN-13: 978-0723434115

Avery, JK. Essentials of Oral Histology and Embryology- A Clinical Approach. 2nd. Mosby A Harcourt Health Sciences Company. ISBN-0323004601
Or
Chiego, DJ, Jr. Essentials of Oral Histology and Embryology- A Clinical Approach. 6th. Mosby A Harcourt Health Sciences Company. ISBN-978-0-3234-9725-1
2. Histology Lecture PowerPoint files (lectures are provided by faculty members)
3. Reference Information (includes descriptive terms, histology stains, histological terminology, tissue preparation, a list of slides and classification of glands).
4. Online Quizzes

Useful websites:

<http://medcell.med.yale.edu/histology/histology.php>

<https://histology.medicine.umich.edu/full-slide-list>

<http://www.adea.org>

<https://histologyguide.org>

<https://umc.libguides.com/sod/DENT601AB>

Course Schedule:

Sessions	Date	Time	Presenter	Activity	Topic	Room	Reading assignments
1	Mon 8/17/26	8am-8:30am		Self-study	Cell Biology		Chapter-Cell (cell membrane, nucleus & organelles)
		8:30am-9:00am	Dr. Cui	Lecture	Intro histology to DENT 601	SM 326	
		9am-9:50am	Dr. Cui	Lecture	Cell Biology I: Cell Membrane, Nucleus & Organelles	SM 326	
		10am-10:50am	Dr. Cui	lecture	Cell Biology II: Protein Synthesis, Cytoskeleton, Cell Structure & Function		
		11am-11:50am	Dr. Cui Staff	Lab	Histology Laboratory introduction/ Meet your workstation	SM 326	
2	Wed 8/19/26	8am-8:50am		Self-study	Epithelium		Chapter-Epithelium
		9am-9:50am	Dr. Cui	Lecture	Basic Tissue 1: Epithelial Tissue	SM 326	
		10am-11:50am	Staff	lab	Classify epithelium & ID structures & cells	SM 326	
3	Mon 8/24/26	8am-8:50am		Self-study	Secretory Epithelium & Exocrine glands		Chapter-Secretory Epithelium & Exocrine Glands
		9am-9:50am	Dr. Cui	Lecture	Basic Tissue 1: Secretory Epithelium & Exocrine glands	SM 326	
		10am-11:50am	Staff	lab	Classify type of exocrine glands, ID structures & cells	SM 326	

4	Wed 8/26/26	8am-8:50am		Self- study	Connective Tissue		Chapter- Connective Tissue
		9am-9:50am	Dr. Cui	Lecture	Basic Tissue 2: Connective Tissue	SM 326	
		10am-11.50am	Staff	Lab	Connective Tissue: Classify the type of CT and fibers. ID CT cells	SM 326	
5	Mon 8/31/26	8am-8:50am		Self- study	Cartilage & Bone		Chapter-Cartilage and Bone
		9am-9:50am	Dr. Cui	Lecture	Basic Tissue 3: Cartilage and Bone	SM 326	
		10am-11:50am	Staff	Lab	Cartilage and Bone: Classify type of cartilage & bone; ID Cells & structures	SM 326	
6	Wed 9/2/26	8am-8:50am		Self- study	Muscle Tissue		Chapter Muscle
		9am-9:50am	Roberts	Lecture	Basic Tissue 4: Muscle Tissue	SM 326	
		10am-11:50am	Staff	Lab	Classify MT; ID cells & structures	SM 326	
7	Wed 9/9/26	8am-8:30am		Self- study	Nervous Tissue		Chapter Nervous Tissue
		8:30am-9:30am	Dr. Cui	Lecture	Basic Tissue 5: Nervous Tissue- PNS & CNS	SM 326	
		9:40am-11:00am	Staff	Lab	Nervous Tissue: Peripheral nervous system (PNS) & Central nervous system (CNS)	SM 326	
		11: 10am - 11:50am	Dr. Cui	Lab	Lab Basic Tissue Review		

8	Mon 9/14/26	8:30am-9:45am		EXAM 1	Lab Practical Exam: Sessions 2 to 7	SM 326	
		10:05am- 11:50am		EXAM 1	Written Exam (MCQ): Sessions 1 to 7	SM 326	
9	Wed 9/16/26	8am-8:25am		Self- study	Peripheral Blood & Blood Development		Chapter peripheral blood & blood development
		9am-9:50am	Dr. Cui	Lecture	Peripheral Blood & Blood Development	SM 326	
		10am-11:50am	Staff	Lab	Peripheral Blood	SM 326	
10	Mon 9/21/26	8am-8:25am		Self- study	Circulatory system		Chapter Circulatory system
		9am-9:50am	Dr. Cui	Lecture	Circulatory System	SM 326	
		10am-11:50am	Staff	Lab	Circulatory system	SM 326	
11	Wed 9/23/26	8am-8:50am		Self- study	Immune System & Lymphoid Organs		Chapter Immune System & Lymphoid Organs
		9am-9:50am	Dr. Cui Dr. Cui	Lecture Lecture	Immune System Lymphoid Organs	SM 326	
		10am-11:50am	Staff	Lab	Lymphoid Organs	SM 326	
12	Mon 9/28/26	8am-8:50am		Self- study	Integument-Skin		Chapter Integument-Skin
		9am-9:50am	Dr. Cui	Lecture	Integument-Skin	SM 326	
		10am-11:50am	Staff	Lab	Integument-Skin	SM 326	
13	Wed 9/30/26	8am-8:50am		Self- study	Respiratory System		Chapter Respiratory System
		9am-9:50am	Dr. Cui	Lecture	Respiratory System	SM 326	
		10am-11:50am	Staff	Lab	Respiratory System	SM 326	

14	Mon 10/5/26	8am-8:50am		Self-study	Urinary System		Chapter Urinary System
		9am-9:50am	Dr. Cui	Lecture	Urinary System	SM 326	
		10am-11:50am	Staff	Lab	Urinary System	SM 326	
15	Wed 10/7/26	8am-8:30am		Self-study			Chapter peripheral blood, circulatory, immune/lymphoid, skin, respiratory & Urinary system
		8:30am-11:30am	Dr. Cui, Dr. Daley Dr. Meyer Dr. Yang	Multiple Discipline Review Lectures	Organ System 1 Review & Clinical Correlations	SM 326	
		11:40am-12:00pm	Staff	Lab	Lab review	SM 326	
16	Mon 10/12/26	8:30am-9:45am		EXAM 2	Lab Practical Exam: Sessions 9 to 15	SM 326	
		10:05am-11:50am		Exam 2	Written Exam (MCQ): Sessions 9 to 15	SM 326	
17	Wed 10/14/26	8am-8:50am		Self-study	Endocrine System		Chapter Endocrine System
		9am-9:50am	Dr. Cui	Lecture	Endocrine System	SM 326	
		10am-11:50am	Staff	Lab	Endocrine System	SM 326	
18	Mon 10/19/26	8am-8:50am		Self-study	Digestive Tract		Chapter Digestive Tract
		9am-9:50am	Dr. Cui	Lecture	Digestive Tract	SM 326	
		10am-11:50am	Staff	Lab	Digestive Tract	SM 326	
19	Mon 10/26/26	8am-8:50am		Self-study	Digestive Glands		Chapter Digestive Glands
		9am-9:50am	Dr. Cui	Lecture	Digestive Glands & Associated Organs:	SM 326	

		10:00am-10: 30	Dr. Cui		liver, pancreas and gallbladder Salivary glands		
		10:30am-11:50am	Staff	Lab	Digestive Glands: glands, liver, pancreas and gallbladder Salivary glands: Parotid, submandibular and sublingual glands	SM 326	
20	Monday 11/2/26	8am-8:50am		Self-study	Oral Embryology: Development of the Oral Facial Region & Face and Palate		Chapters Development of the Oral Facial Region & Face and Palate (Oral Histology & Oral Embryology)
		9am-9:50am	Dr. Meyer	Lecture	Embryology 1- Development of the Facial Region: Trilaminar disk formation and Pharyngeal arches	SM 326	
		10am-10:50am	Dr. Meyer	Lecture	Embryology 2-Oral cavity, palate, nasal cavity and tongue	SM 326	
		11am-11:50am	Staff	Lab	Oral Embryology	SM 326	
21	Wed 11/4/26	8am-8:30am		Self-study			Chapters Urinary system, Endocrine system, Digestive tract, Digestive Glands & Oral Embryology
		8:30am-11:30am	Dr. Cui Dr. Daley Dr. Meyer Dr. Yang	Multiple Discipline Review Lectures	Organ System 2 & Embryology Review & Clinical Correlations	SM 326	

		11:40am-12:00am	Staff	Lab	Lab review	SM 326	
22	Mon 11/9/26	8:30am-9:45am		EXAM 3	Lab Practical Exam: Sessions 17 to 21	SM 326	
		10:05am-11:50am		Exam 3	Written Exam (MCQ): Sessions 17 to 21	SM 326	
23	Wed 11/11/26	8am-8:50am	Dr. Cui	Lecture	Oral mucosa and Tongue	SM 326	Chapter Oral mucosa and Tongue
		9am-9:50am	Dr. White	Lecture	Correlation: Oral pathology	SM 326	
		10am-11:50am	Staff	Lab	Oral mucosa and Tongue	SM 326	
24	Mon 11/16/26	8am-8:50am		Self-study	Odontogenesis		Chapters Development of teeth, eruption & shedding of teeth (Oral Histology & Embryology)
		9am-9:50am	Dr. Cui	Lecture	Odontogenesis	SM 326	
		10am-11:50am	Staff	Lab	Odontogenesis	SM 326	
25	Wed 11/18/26	8am-8:50am		Self-study	Enamel, Dentin & Pulp		Chapters Enamel, Dentin & Pulp (Oral Histology & Embryology)
		9am-9:50am	Dr. Cui	Lecture	Enamel, Dentin & Pulp	SM 326	
		10am-11:50am	Staff	Lab	Enamel, Dentin & Pulp	SM 326	
26	Mon 11/30/26	8am-8:50am		Self-study	Periodontium, Cementum, PDL & alveolar bone		Chapters Periodontium, Cementum, PDL & alveolar bone (Oral Histology & Embryology)

		9am-9:50am	Dr. Cui	Lecture	Periodontium, Cementum, PDL & alveolar bone	SM 326	
		10am-11:50am	Staff	Lab	Periodontium, Cementum, PDL & alveolar bone	SM 326	
27	Wed 12/2/26	8am-8:30am		Self-student			Chapter 13 Temporomandibular joint (Oral Histology & Embryology)
		8:30am – 9:00am	Lab	Dr. Cui	Lab Oral Cavity Review	SM326	
		9am-9:50am	Dr. Brain	Lecture	Clinical Correlation: Periodontium	SM 326	
		10am-10:50am	Dr. Bain	Lecture	Clinical correlation: Osseo-integration, implants	SM 326	
		11am-11:50am	Dr. Holesch	Lecture	TMJ and clinical correlation		
28	Thurs 12/10/26	8:30am-9:45am		EXAM 4	Lab Practical Exam: Sessions 23 to 27	SM 326	
		10:05am-11:50am		Exam 4	Written Exam (MCQ): Sessions 23 to 27	SM 326	

Class Policies and Procedures

1. Attendance Policy: SOD Attendance Policy and SOD Non-Clinical Dress code policy (from student Handbook) is posted in Syllabus in Canvas and will be reviewed at the first day of class session.
2. DENT 601A Assessment and Grading Policy is posted in Canvas and will be reviewed at the first day of class session.

SOD Student Handbook link:

<https://umc.edu/sod/Students/Current%20Students/SOD-Student-Affairs/SOD-Student-Handbook.html>